

GRA-Health-Swarm: Swarm Subjectivity for Health, Rejuvenation and Stress Resilience via Anokhin's Functional Systems and GRA Nullification

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Abstract

This paper presents a mechanism for cybernetic modeling of **health, stress and disease** through P.K. Anokhin's Theory of Functional Systems combined with GRA-nullification. The **GRA-Health-Swarm** repository implements this mechanism in software.

Key idea:

- **Disease** = prolonged mismatch between the brain's prediction (Acceptor of Action Result – AAR) and reality.
- In GRA terms: disease = chronically high foam $\Phi_{\text{total}} \gg 0$.
- **Health** = regime $\Phi_{\text{total}} \rightarrow 0$ via GRA-reset at the swarm level.

A swarm of agents models functional subsystems of the organism (stress, goals, body, environment, threats). The $\mathbf{N}_{\text{health}}$ operator (GRA-reset adapted for health) reconfigures goals, expectations and interactions, striving for a coherent state.

Scenarios:

- **drone_phobia/** – constant fear of Ukrainian drones, chronic mismatch.
- **burnout/** – work/war burnout.
- **rejuvenation/** – rejuvenation and longevity.
- **chronic_disease/** – progression of chronic disease.

Rejuvenation theorem: if a biological swarm has at least one agent with a correct AAR (prediction of healthy state) and the $\mathbf{N}_{\text{health}}$ operator is applied iteratively, then $\Phi_{\text{total}} \rightarrow 0$ is achievable.

This is **not a medical tool**, but a cybernetic model: a system can survive war without internal collapse.

1 Introduction

In modern approaches to health and stress, often individual aspects are considered: medicine, psychology, physiology. Within the GRA (Gradient Reduction of Argumentative Foam) theory and Anokhin's Theory of Functional Systems, the notion of **foam** Φ is introduced as a measure of internal uncertainty, contradictoriness and mismatch.

Academician P.K. Anokhin showed that:

1. The brain builds a **prediction of the result** (Acceptor of Action Result – AAR).
2. If reality **matches** the prediction – stability, no stress.
3. If reality **does not match** and the mismatch persists – **chronic stress** $\rightarrow$ **disease**.

This work proposes a mechanism where a swarm of agents (functional systems) can maintain itself in a state of **health** through foam Φ nullification, even under war, drone attacks and chronic stress.

2 Theoretical Foundation

2.1 Anokhin’s Theory of Functional Systems

Definition 1. A *functional system* is an organization of activity of elements of different anatomical membership, directed towards achieving a useful adaptive result [Anokhin].

Key components:

- **Afferent synthesis** – gathering information from the environment.
- **Dominant** – main goal.
- **Acceptor of Action Result (AAR)** – prediction of future result.
- **Efferent action** – motor / physiological output.
- **Reverse afferentation** – feedback from body and environment.
- **Result – mismatch** – error signal when prediction and reality do not match.

2.2 GRA and Foam

In GRA theory:

Definition 2. *Foam* Φ of an agent is a measure of internal contradictoriness of its argumentative space. High Φ corresponds to uncertainty, blurriness of goals and internal instability.

Definition 3. *Health* $\approx$ low Φ_{total} (coherent functional system).

Definition 4. *Disease / burnout / drone-phobia* $\approx$ chronically high Φ_{total} (chronic mismatch).

2.3 The Anokhin–GRA Bridge

Table 1: Mapping Anokhin’s components to GRA

Anokhin	GRA
AAR (result prediction)	Agent’s world model Ψ
Reality	Data from environment
Mismatch	Foam $\Phi > 0$
Chronic mismatch	Prolonged high Φ
Stress	System instability
Disease	System “sick” = Φ not minimized
Recovery	GRA-reset: operator $N \rightarrow \Phi \rightarrow 0$

Theorem 1 (Disease and Health in GRA). *Disease* = prolonged mismatch between brain’s prediction and reality.

Healing = swarm-level GRA-reset pushing $\Phi_{total} \rightarrow 0$.

3 GRA-Reset for Health

3.1 The N_{health} Operator

The N_{health} operator (GRA-reset adapted for health):

1. Computes local foam Φ_i of each agent (stress, body, goals, environment).
2. Detects high-foam configurations (chronic mismatch).
3. Reconfigures goals, expectations and interactions:
 - Adjusts unrealistic predictions (AAR recalibration).
 - Strengthens feedback loops (reverse afferentation).
 - Reduces contradictory goals between agents.
 - Activates compensatory functional systems.
4. Strives for $\Phi_{\text{total}} \rightarrow 0$.

Theorem 2 (Rejuvenation Theorem). *If a biological swarm has at least one agent with a correct AAR (prediction of healthy state) and the N_{health} operator is applied iteratively, then $\Phi_{\text{total}} \rightarrow 0$ is achievable, corresponding to a state of optimal health and rejuvenation.*

4 Implementation

The GRA-Health-Swarm repository implements the described mechanism:

- **core/** – agents (stress, body, goals, environment), N_{health} operator, interaction graph.
- **scenarios/** – demonstration runs:
 - **drone_phobia/** – constant fear of Ukrainian drones.
 - **burnout/** – work/war burnout.
 - **rejuvenation/** – rejuvenation and longevity.
 - **chronic_disease/** – progression of chronic disease.
- **math/** – formal description of the theory.
- **paper/** – this paper.

5 Scenario: Drone-Phobia

Prediction: “I am safe, I am at home, there are no threats”.

Reality: constant drone attacks, explosions, alarms, sirens [BBC, New Izvestia].

Mismatch $\rightarrow$ stress $\rightarrow$ drone-phobia.

In GRA:

- Agent “threat” $\rightarrow$ high Φ .
- Agent “safety” $\rightarrow$ low Φ only when there is shelter.
- Φ_{total} remains high $\rightarrow$ system “sick”.

What can be done:

- Reduce actual threat (fewer drones).
- Increase shelter availability.
- Reduce unpredictability (warnings, signals).
- Change environment (safe zones).

Operator N_{health} does not stop war, but helps **not to lose one's mind in it**.

6 Discussion

The proposed mechanism shows that:

- Disease can be modeled as chronically high foam Φ .
- Health is the regime $\Phi_{\text{total}} \rightarrow 0$ via GRA-reset.
- “Healing” is not “imagine yourself healthy”, but structural reorganization of a multi-agent functional system.
- This is **not a medical tool**, but a cybernetic model.
- A system can survive war without internal collapse if $\Phi_{\text{total}} \approx 0$ is maintained.

Future work:

- Formalize the Rejuvenation Theorem in rigorous mathematical form.
- Investigate health stability against disruption of links in the swarm.
- Test the mechanism on real multi-agent systems.

7 Connection to GRA Ecosystem

- GRA-Multiverse-Final – universal operator N ($\Phi \rightarrow 0$).
- GRA-Swarm-Subject – subjectivity contagion in swarm.
- GRA-Health-Swarm – specialization on health: “subject” = healthy organism, “swarm” = its functional systems.

Important Disclaimer

This repository is NOT a medical tool. It is a cybernetic model of health, stress and resilience:

- It **does not stop wars** or eliminate drones.
- It **does not replace medical treatment**.
- It **explains the mechanism**: why chronic mismatch causes disease and how structural reorganization can restore coherence.

Acknowledgments

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License

The project code is released under the MIT License. The full license text is included in the `LICENSE.txt` file of the repository.